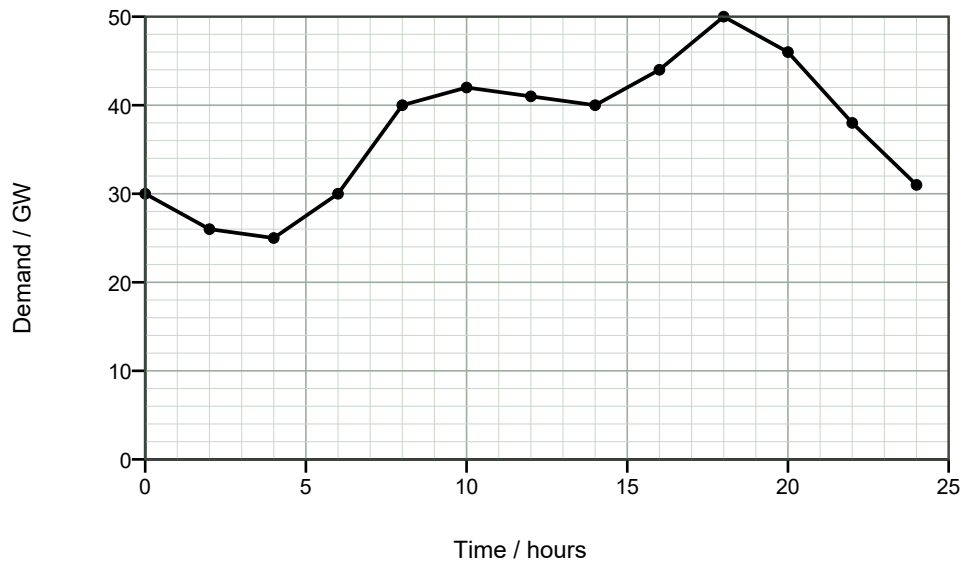
# Questions by topic: Fluctuations in supply/demand & energy storage - Set 1 (60 marks)

**Answer all questions in the spaces provided.**

**0 1** Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

**Figure 1**




**0 1 . 1** Describe the pattern of demand shown in Figure 1.

**[2 marks]**

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**0 1 . 2** Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

**[4 marks]**

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**0 1 . 3** Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

**[2 marks]**

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**0 1 . 4** Give **two** causes of fluctuations in energy **supply**.

**[2 marks]**

**1** .....

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**2** .....

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**10**

**0 2** A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million $m^3$ of water. The density of water is 1000 kg $m^{-3}$ and g = 9.8 N $kg^{-1}$.

**0 2 . 1** Describe how pumped-storage HEP is used for peak shaving.

**[3 marks]**

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**0 2 . 2** Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

**[3 marks]**

**Answer** ☐ J

**0 2 . 3** The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

**[2 marks]**

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**0 2 . 4** Give **two** locational requirements of a pumped-storage station.

**[2 marks]**

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**2** .....

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**0 3 . 1** Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

**[4 marks]**

**0 3 . 2** Compare **three** methods of storing hydrogen, giving a disadvantage of each.

**[3 marks]**

**0 3 . 3** Explain how a fuel cell differs from a rechargeable battery.

**[2 marks]**

**0 3 . 4** Explain what is meant by the 'hydrogen economy'.

**[1 mark]**

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**0 4** Table 1 shows features of three rechargeable battery types.

**Table 1**

| Battery type | Energy density / Wh kg$^{-1}$ | Charge-discharge cycles | Storage efficiency / % | Cost per kWh / £ | Safety issues |
|---|---|---|---|---|---|
| Lead-acid | 35 | 500 | 80 | 100 | Toxic lead, acid |
| Nickel-cadmium | 50 | 1500 | 75 | 250 | Toxic cadmium |
| Lithium-ion | 200 | 2000 | 92 | 150 | Fire risk |

**0 4 . 1** Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

**[4 marks]**

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**0 4 . 2** Give **two** other factors that affect the viability of a rechargeable battery.

**[2 marks]**

**1** .....

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**2** .....

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**0 4 . 3** Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

**[2 marks]**

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**0 4 . 4** Explain how Power-to-Gas (P2G) systems use existing infrastructure.

**[2 marks]**

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**0 5 . 1** Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

**[2 marks]**

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**0 5 . 2** Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

**[2 marks]**

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**0 5 . 3** Explain why molten salt rather than water is used to store heat from CSP plants.

**[2 marks]**

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**0 5 . 4** Explain how high thermal mass building materials reduce energy use.

**[2 marks]**

**0 5 . 5** Describe **one** other energy storage technology.

**[2 marks]**

**10**

**0 6 . 1** Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

**[9 marks]**

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**0 6 . 2** State what is meant by **peak shaving**.

**[1 mark]**

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**10**

**END OF QUESTIONS**

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